

Forecasting and Valuing Cash Flows

From Chapter 2 of *Valuation: The Art and Science of Corporate Investment Decisions*, 2/e.
Sheridan Titman. John D. Martin. Copyright © 2011 by Pearson Education. Published by
Prentice Hall. All rights reserved.

Forecasting and Valuing Cash Flows

Chapter Overview

Discounted cash flow (DCF) analysis is a key building block for a valuation course. This chapter, which introduces the process for applying DCF analysis, describes the relationship between cash flows and numbers from the firm's financial statements. We first define cash flow to the equity holders and then cash flow available for distribution to both creditors and owners. After defining cash flows, we present a comprehensive cash flow forecasting example that focuses on the process that one goes through to link units sold, unit costs, and unit revenues to investment cash flows. Finally, the chapter introduces the mechanics of discounting investment cash flows to estimate project value.

1 DISCOUNTED CASH FLOWS AND VALUATION

The idea behind **discounted cash flow (DCF) valuation analysis** is simple: The value of an investment is determined by the magnitude and the timing of the cash flows it is expected to generate. The DCF valuation approach provides a basis for assessing the value of these cash flows, and consequently it is a cornerstone of financial analysis.

How does DCF analysis fit into the overall investment evaluation process laid out in the previous chapter? You may recall that we discussed the investment valuation process in terms of three phases of analysis that can be broken down into eight key steps. The focus of this chapter is on Step 2 of the first phase—estimating the investment's value, or “crunching the numbers.”

The popularization of DCF for project analysis is generally attributed to Joel Dean (1951). However, the roots of DCF analysis go back much further. For example, financial mathematics can be traced back to the early writing of Leonardo of Piza (also known as Fibonacci) in 1202. Generations of business school graduates have made DCF one of the most widely used tools in the arsenal of today's finance professional.

Example—Car Wash

To illustrate the role of DCF valuations in analyzing an investment opportunity, let's assume that you invest in a car wash. The car wash generates *cash inflows* each time a

customer purchases a wash and *cash outflows* when you purchase supplies, pay taxes and wages, and so forth. In addition, there will be times when you have to make repairs to the equipment, resulting in nonroutine cash outflows.

Let's now assume that you open a bank account for the business, and the cash receipts flow into the account and cash disbursements come out of it. After a year of operations, you have built up a cash balance of, say, \$10,000 after paying all your bills (including any repairs to your equipment). If we assume for the moment that you have not borrowed any of the money used to invest in the car wash, then you have \$10,000 that you can distribute to yourself as sole owner of the car wash. This is money you can spend right now, at the end of the first year of operations. Looking forward to years two, three, and so forth and performing a similar computation, you can determine future cash flows—calculated after paying all the firm's operating expenses and making any added capital expenditures. These cash flows are the key element that determines the value of the business.

The Three-Step DCF Process

From this simple example of the car wash, we can develop a DCF process that takes into account the timing of cash flows. Specifically, the use of DCF valuation entails a three-step process, as depicted in Figure 1: The first step involves forecasting the amount and timing of the anticipated receipt (or payment) of future cash flows. Here the basic issue is: *How*

Did you know?

Forecasting is tough: Even the experts have trouble spotting market trends.

Forecasting the future is difficult, especially when it comes to new technology, as the following quotes from the Hall of Forecasting Shame will attest:

“I think there is a world market for maybe five computers.”

— Thomas Watson, Chairman of IBM, 1943

“There is no reason anyone would want a computer in their home.”

— Ken Olson, president, chairman, and founder of Digital Equipment Corporation, 1977

“640K ought to be enough for anybody.”

— Bill Gates, 1981

Figure 1 Steps in Performing a Discounted Cash Flow Analysis

STEPS	INVESTMENT VALUATION
Step 1: Forecast the amount and timing of future cash flows. <i>How much cash is the project expected to generate and when?</i>	Forecast free cash flow (FCF).
Step 2: Estimate a risk-appropriate discount rate (covered in Chapters 4 and 5). <i>How risky are the future cash flows, and what do investors currently expect to receive for investments of similar risk?</i>	Combine the debt and equity discount rate (weighted average cost of capital, WACC).
Step 3: Discount the cash flows. <i>What is the present value “equivalent” of the investment’s expected future cash flows?</i>	Discount FCF using WACC to estimate the value of the project as a whole.

much cash is the project *expected* to generate and *when*? The real “heavy lifting” in project evaluation occurs here, for the future can never be known with certainty, and forecasts must often be made with little or no historical data to guide the analysis.

Second, the analyst must determine the appropriate discount rate for use in discounting future cash flows back to the present. The fundamental issue here is: *How risky are the future cash flows*? This topic is a central focus of finance. For now, we will simply assume that discount rates are known.

Finally, Step 3 involves the mechanical process of discounting future cash flows back to the present. Although we leave the rudiments of this analysis to more introductory finance textbooks,¹ we do offer a brief review in Section 4.

2 DEFINING INVESTMENT CASH FLOWS

Arriving at an estimate of the value of an investment using DCF analysis requires that the analyst have a good understanding of the investment’s cash flows. In this section, we discuss three key issues related to the proper definition of investment cash flows:

1. What cash flows are relevant to the valuation of a project or investment?
2. Are the cash flow forecasts conservative or optimistic?
3. What is the difference between equity and project cash flows?

Failure to address each of these issues has led to a great deal of confusion in the application of DCF analysis.

Relevant Cash Flows

By “relevant” we mean that the cash flows have an impact on the valuation of the project under consideration. There are two important questions we need to consider when evaluating whether a cash flow is relevant or not. First, are the cash flows incremental to the project under consideration? Second, do the cash flows represent sunk costs? We consider each of these questions in turn.

Only Incremental Cash Flows Are Relevant

Only cash flows that are a direct result of the acceptance of an investment are relevant to the valuation of the project. These are often referred to as incremental cash flows, because they are the additional cash flows to the firm that are generated by the investment. These include the cash flows directly generated by the investment as well as the indirect effects that the investment may have on the firm’s other lines of business. For example, when Frito-Lay evaluates the introduction of a new flavor of Doritos brand Tortilla Chips (e.g., lime flavored), the projected revenues and costs of the new product are critical to the analysis. However, the extent to which the new product’s sales may cannibalize, or source

¹ For example, Sheridan Titman, Arthur J. Keown and John D. Martin, *Financial Management: Principles and Applications*, 11th ed. (Upper Saddle River, NJ: Pearson Prentice Hall, 2011), Chapter 5.

sales from, other existing products (such as Doritos Nacho Cheese–flavored tortilla chips and Doritos Cool Ranch–flavored tortilla chips) is equally critical, because the true net or incremental cash flows are the new-product cash flows net of cannibalization. See the Practitioner Insight box for a look at Frito-Lay’s three-step incremental cash flow analysis.

Sunk Costs Are Not Relevant

A common mistake in the calculation of incremental cash flows has to do with what are known as *sunk costs*. Sunk costs are expenditures that either have already been made or must be made regardless of whether the firm proceeds with the investment. As a result, sunk costs are not incremental costs and should thus be ignored in the investment analysis.

For example, suppose that Merck previously invested \$10 million in the research and development of a new methodology for extracting stem cells from adult donor cells that does not require the use of embryos. The procedure looked very promising for the development of cancer treatments; however, it failed to deliver any significant improvements over less expensive and more conventional treatments during clinical trials. Suppose that subsequent Merck researchers develop a way to apply the new

PRACTITIONER INSIGHT

Relevant Cash Flows and Revenue Cannibalization

Frito-Lay, one of four divisions of PepsiCo Inc., is currently the market leader for potato chips and other salty snack products. Consequently, when Frito-Lay evaluates a new salty snack product, they realize that a portion of the new product sales will come from the lost sales from existing products (i.e., revenue cannibalization). Such a challenge is common in mature firms with large market share. The evaluation of incremental cash flows is thus an extremely important issue for Frito-Lay and has led the company to develop a formal approach to the incremental effects of new product offerings on the firm’s cash flows.

At Frito-Lay, they use the following three-step approach when estimating incremental cash flows:

- Step 1:** Estimate the total revenue that will be generated by the product.
- Step 2:** Estimate what percent of the revenue is true incremental revenue.
- Step 3:** Estimate the incremental cash flow.

To estimate the percent of the revenue that is incremental, we classify new product offerings

into one of three types. The first type contains projects that have a high likelihood of cannibalizing existing product sales. Examples of this would include products such as a new flavor of Doritos brand Tortilla Chips. The expectation here is that a low percentage of the sales of this type of product can be viewed as incremental sales from the product, with the remainder coming from reductions in existing product sales. The second type of products provides the potential for additional store shelf space because they provide some new benefit (e.g., baked chips). For this type of product, Frito-Lay assigns a higher percentage of the revenue as true incremental sales. Finally, the third type of product, such as Frito-Lay’s natural line, provides the opportunity to enter new channels and/or develop new shelf space in a different part of the store. These products carry the lowest risk of cannibalization and are therefore assigned the highest percentage of incremental sales.

*Based on interview with Keith Crider, new product finance manager, Frito-Lay, Dallas, TX.

Did you know?**Men tend to be more overconfident than women.**

Psychologists have shown that men tend to be more confident than women in a number of settings, including those involving financial forecasts. A study by Barber and Odean* shows that among individual investors, men tend to trade more than women (showing more confidence) but perform worse!

*Brad M. Barber and Terrance Odean, "Boys Will Be Boys: Gender Overconfidence and Common Stock Investment," *The Quarterly Journal of Economics* (February 2001).

procedure to more efficiently carry out embryo transplants in beef and dairy cattle. In calculating the value of an investment associated with commercializing this procedure, how should Merck treat the original \$10 million R&D expenditure?

In general, the research and development costs (incurred in the past) should be viewed as sunk costs and should not be relevant to the analysis of the value of marketing the process for the cattle embryo-transplant project. However, the past research would become relevant if Merck receives an offer from Pfizer to acquire the technology for \$8 million under the proviso that Merck give up all rights to the further development and applications of the technology. In this instance, the relevant cost to Merck of using the technology to develop its application for the cattle embryo-transplant market is now equal to the \$8 million offer from Pfizer (assuming this is the highest bid received), not the \$10 million Merck invested in R&D to develop the methodology. The reason for this is that the original \$10 million has been spent and cannot be recovered. However, if Merck proceeds with plans to apply the technology to the embryo-transplant market, it must forgo the opportunity to sell the technology to Pfizer for \$8 million. Clearly, the \$8 million represents an opportunity cost to Merck.²

The determination of relevant cash flows can sometimes be very difficult, as we have just illustrated. However, it is critical that only the incremental revenues and incremental costs that are a direct result of the firm's decision to undertake the investment be considered in performing the valuation of the project.

Conservative and Optimistic Cash Flows

When academics talk about valuing an investment by discounting cash flows, they generally assume that the cash flows represent "expected cash flows." In the statistical sense, they assume that managers estimate the cash flows that the firm *expects* to realize in various scenarios and sum these cash flows after weighting them by their probabilities of occurrence. For example, if a firm's manager expects to generate a cash flow of either \$50,000 or \$100,000, each with a 50% probability, the expected cash flow is \$75,000. In theory, the firm should discount these expected cash flows, as we noted in Figure 1, using a risk-adjusted rate of interest that reflects the risk of the cash flows. In practice, however, the cash flow forecasts that managers use are frequently not the same as the expected cash flows that the academics describe in their theories. Depending on the situation, the cash flow forecasts can be biased as a result of managers being either too conservative or too optimistic. For example, managerial bonuses are sometimes tied to getting new projects approved. Such an incentive might encourage managers to be very aggressive in their cash flow forecasts. In addition, managers may be overly optimistic in their forecasts simply because of overconfidence in their own analyses. (See the

² One easy way to trap incremental costs is to take the difference between cash flows that would arise if the project is taken up and the free cash flows that would arise if the project is rejected. For instance, the \$10 million R&D expense would occur in either case (project taken up or rejected), whereas the \$8 million opportunity loss (cost) would arise only if the project is rejected (and the technology is sold to Pfizer).

Behavioral Insight entitled “Overconfidence.”) In either of these cases, the cash flow forecasts would be biased upward. Managerial forecasts can also be biased downward. This might occur when managerial pay is tied to meeting and exceeding the cash flow forecasts. In this instance, managers may bias their cash flow estimates downward to increase the likelihood that they will be able to meet and exceed the cash flow forecast.

In order to understand how managers make cash flow forecasts in practice, it is useful to reconsider the investment process and to consider the incentives of the various players who are involved in making the forecasts. Suppose, for example, that you are proposing an investment that you will be managing and that your cash flow forecasts will serve as future targets that will influence your future bonuses. If this is the case, then you might choose relatively conservative forecasts. Now consider a situation where you get a bonus for identifying a promising investment opportunity that the firm initiates. When this is the case, you might choose relatively optimistic forecasts.

We believe that top executives may encourage their managers to develop forecasts that represent *hoped-for* rather than expected cash flows, because these forecasts provide future targets that may serve as motivation when the project is implemented. We also observe management cash flow forecasts that are based on the reasoning: “If all goes as planned, these are the cash flows that we expect to achieve.” These optimistic cash flow forecasts ignore various unanticipated glitches that may arise when the project is implemented. When evaluating hoped-for cash flows, firms should use very high discount rates to adjust for the difference between dreams and reality.

Two examples where we tend to observe what are clearly hoped-for rather than expected cash flows are in emerging markets and venture capital investments. In both cases, the firm may have a strategic partner—i.e., the government of a developing country in one case and an entrepreneur in the other case—who prefers a business plan that

BEHAVIORAL INSIGHT

Overconfidence

Psychological studies show that most people are overconfident about their abilities and tend to be optimistic about the future. For example, one study found that most people claim to be better-than-average drivers, yet we know that by definition half are above average and half are below average.* Are corporate executives likely to be more or less optimistic and overconfident than the average individual? We would argue that they are likely to be more optimistic and overconfident. Psychologists claim that individuals tend to be subject to what they call self-attribution bias, which means that they want to attribute good things that happen to them to their own efforts and ability. This suggests that top

executives are likely to attribute their success to their own abilities, rather than to being in the right place at the right time, and may thus be overconfident about their success going forward. It’s also likely to be the case that the most optimistic managers work the hardest (because they place a high value on being promoted), which suggests that optimistic people are more likely to be promoted. What this means is that even well-intentioned executives who set out to estimate true expected cash flows may end up with cash flow forecasts that are overly optimistic.

*Ola Svensson, “Are We All Less Risky and More Skillful than Our Fellow Drivers?” *Acta Psychologica* 47 (1981): 143–48.

is not so specific about possible negative events that could occur. For example, in a proposed joint venture with the government of Venezuela, you might not want to explicitly account for the possibility of a collapse in the current regime. You may prefer to calculate hoped-for cash flows and account for the risk of collapse with an adjustment in the required discount rate. Similarly, as a venture capitalist, you may not want to explicitly express a lack of confidence in the viability of the entrepreneur's new product, but may again adjust for the optimistic, hoped-for cash flows by requiring a very high discount rate to adjust for the difference between dreams and reality. For the balance of this chapter, we will assume that all cash flow estimates represent *expected* cash flows.

Equity Versus Project Free Cash Flow

Keep in mind that an investment project's cash flow is simply the sum of the cash inflows and outflows from the project. However, as we have already noted, analysts typically structure their analysis of investment cash flows using projected financial statements, commonly referred to as pro forma statements, for the project or firm being valued. That is, they develop their cash flow analysis by first projecting the income or earnings consequences of the project (following accrual accounting—see the Technical Insight box) and then use this information to calculate the project's cash flows.

There are two commonly used definitions of cash flow, and we will use both at different times in this book. We refer to the first simply as free cash flow (FCF). When discussing project valuation, we often use the adjective *project* before *free cash flow*, or **project FCF**. Later, when we discuss the valuation of a business enterprise, we use the term **firm FCF**. In either case, the definition of cash flow is the same. Specifically, FCF is the amount of cash

TECHNICAL INSIGHT

Accrual and Cash-Basis Accounting

Accounting *income* is not the same as cash flow, because it is calculated using the accrual basis of accounting. Unlike cash accounting, **accrual accounting** distinguishes between the recording of costs and benefits associated with economic activities and the actual payment and receipt of cash. For example, under accrual accounting, revenues are recognized (recorded) *when earned*, not when cash actually changes hands. Likewise, under accrual accounting, expenses are matched to the revenues they helped generate, rather than being recorded when cash is actually paid.

Why accrual rather than cash accounting?
Investors' demands for *periodic financial reports*

that can be used to assess performance gave rise to the need for the accrual system. The problem with cash-basis accounting is that cash expenditures made in one period may have an impact on revenues and profits over multiple future periods. Accrual accounting is thus designed to match expenses with revenues. For example, the cost of a piece of capital equipment that will be productive over many years is spread over multiple years, rather than being expensed entirely in the period of the purchase. Cash-basis accounting would expense the equipment cost, which would distort the performance of the firm's operations during the period.

flow (produced by an individual investment project or an entire firm) that is available for distribution to the various claimants (debt and equity) after paying all the firm's expenses and investing in new projects.

In addition to project and firm free cash flow, there is also the concept of **equity FCF**. Equity FCF refers to the amount of cash flow that is available for distribution to the firm's equity holders. Therefore, equity FCF equals firm or project FCF, minus the sum of the after-tax interest paid to the firm's creditors, plus the net new debt (i.e., new debt issued by the firm less principal repayments) for the period. For now, we focus strictly on firm or project FCF, or simply FCF.

Calculating Project Free Cash Flow (FCF)

It is worth taking a moment to consider what we mean by the term *free* in the context of cash flow. The idea is not that the cash flow is in any sense free of cost. Instead, *free* refers to the fact that the cash flow under discussion is available—not needed for any particular purpose. The cash flow equals the amount of cash left over after paying all expenses, including any additional investments made during the period. The firm, then, can distribute the remaining cash because the firm does not, by definition, need it. Consequently, our cash flow calculations result in a cash flow figure that is free of any encumbrances or commitments and can therefore be distributed back to the sources of capital used to finance the investment. **Free cash flow (FCF)**, therefore, represents the amount of cash produced (by a project or a firm) during a particular period of time that can be distributed to the firm's creditors (principal and interest payments) and stockholders (dividends and share repurchases).

Using a pro forma or planned income statement for a future period, we can compute FCF using the procedure laid out in Figure 2.

Figure 2 Calculation of Free Cash Flow (FCF)

Sales	Note that interest expense is <i>not</i> deducted before calculating the firm's tax liability. This is because FCF represents the cash flow available for both creditors (interest plus principal) and owners. In Chapter 9, where we calculate equity FCF, we will deduct interest expense before calculating the firm's tax liability!
Less: Cost of goods sold	
Gross Profit	
Less: Operating expenses (including depreciation expense)	
Earnings Before Interest and Taxes	
Less: Taxes	Note that from this point, the FCF calculation differs from the normal format of the income statement: (i) depreciation expense is added back to reflect the fact that although tax deductible, it does not represent a cash outflow; (ii) capital expenditures are deducted because they represent a cash outflow; and (iii) increases in net working capital are deducted because they, like capital expenditures for fixed assets, represent a cash outflow for an investment in the firm or project.
Equals: Net operating profit after taxes (NOPAT)	
Plus: Depreciation expense	
Less: Capital expenditures (CAPEX)	
Less: Increases in net working capital (ΔNWC)	
Free cash flow (FCF)	

Note that the calculation of FCF begins with the same elements that are included in a firm's income statement (specifically, a pro forma or planned income statement because we are forecasting future cash flow). The process begins with a prediction of future sales or revenues, the cost of goods sold that are associated with these sales, and the operating expenses incurred in producing the goods and services sold during the period. Deducting cost of goods sold and operating expenses from sales provides an estimate of the firm's operating income, which we refer to as earnings before interest and taxes (EBIT). At this point, the computation of FCF deviates from the income statement in that we compute the firm's tax liability on EBIT without first deducting interest expense, as is done in the income statement. The reason for this is that we are computing the cash flow that will be available for payment to all the firm's sources of financing, including both its creditors (principal plus interest) and equity investors.

Net operating profit after taxes (NOPAT) is then defined as EBIT less taxes, computed using EBIT. Next, we add back depreciation expense to NOPAT, because depreciation is a noncash expense for the period that was deducted in computing EBIT. Finally, we deduct any expenditures the firm has planned for the period to cover the cost of acquiring new capital equipment (CAPEX) and additions to its operating net working capital.

We can also calculate FCF algebraically using Equation 1 as follows:

$$\begin{aligned} \text{Free Cash Flow (FCF) for Year } t = & \left(\frac{\text{Earnings before Interest and Taxes (EBIT) for Year } t}{\text{EBIT for Year } t} \right) \left(1 - \frac{\text{Tax Rate } (T)}{\text{Tax Rate } (T)} \right) \\ & + \frac{\text{Depreciation and Amortization Expense (DA) for Year } t}{\text{DA for Year } t} - \frac{\text{Capital Expenditures (CAPEX) for Year } t}{\text{CAPEX for Year } t} \quad (1) \end{aligned}$$

You will recognize many of the inputs to Equation 1 based on your understanding of a firm's income statement. However, some entries deserve further explanation. We explain these inputs to the FCF definition in the sections that follow.

Depreciation and amortization expense (DA) As you know from your accounting courses, depreciation and amortization expenses do not represent out-of-pocket cash payments. Rather, they arise out of the matching principle of accounting, which dictates that expenses be matched with revenues whenever it is reasonable and practicable to do so. Therefore, firms use *depreciation expense* to match expenditures made for long-lived assets (such as plant, machinery, and equipment) against the revenues the assets help generate. However, the actual expenditure of cash may have taken place many years earlier when the assets were acquired. Thus, *the allocation of the original cost against revenues in the form of depreciation or amortization expenses does not represent an actual cash payment.*

Because the Internal Revenue Service requires that expenditures for long-lived assets be offset (i.e., depreciated or amortized, depending on the type of asset) against revenues throughout their useful lives and because these expenses are tax deductible, we include them in our calculation of free cash flow. First we subtract them in calculating taxable income, and then we add them back to after-tax income to calculate cash

flow. The net result is that we add an amount back to the FCF equal to the tax savings on the depreciation expense.³

Capital expenditures (CAPEX) To sustain a firm's productive capacity and provide for growth in future cash flows, the firm must periodically make investments in new long-lived assets that are typically referred to as property, plant, and equipment. This includes such things as expanding plant capacity and replacing old equipment. These expenditures are referred to as *capital expenditures*, with the acronym *CAPEX*.

CAPEX can be calculated by analyzing how net property, plant, and equipment (net PPE) on the balance sheet changes over time; e.g., consider the change in net PPE from 2009 to 2010.⁴

Net PPE (2009)

Less: Depreciation expense for 2010

Plus: CAPEX for 2010

Equals: Net PPE (2010)

Therefore, CAPEX for 2010 can be calculated as follows:⁵

$$\begin{aligned}\text{CAPEX for 2010} &= \text{Net PPE (2010)} - \text{Net PPE (2009)} \\ &\quad + \text{Depreciation Expense for 2010}\end{aligned}$$

Based on the above relationship, it is obvious that the capital expenditure for a particular year is related to the firm's depreciation expense for the year, but the amounts will not generally be the same. Why are they different? Depreciation expense is determined by a firm's prior expenditures on plant and equipment, which are amortized or expensed against firm revenues over the life of the plant and equipment. The amount of the depreciation expense is determined both by the amount the firm has spent in the past and by accounting (and tax) rules. Capital expenditures, on the other hand, reflect current period expenditures for new property, plant, and equipment.⁶ The amount of CAPEX in any given period reflects the amount of plant and equipment that physically wears out and needs replacement, in combination with the demands of

³ Defining EBITDA as earnings before interest, taxes, and depreciation, or $\text{EBIT} + \text{DA}$, we can restate the FCF formula from Equation 1 as follows:

$$\text{FCF} = (\text{EBITDA} - \text{DA})(1 - T) + \text{DA} - \text{WC} - \text{CAPEX}$$

or, rearranging terms,

$$\text{FCF} = (\text{EBITDA})(1 - T) + T \times \text{DA} - \text{WC} - \text{CAPEX}$$

where $T \times \text{DA}$ represents the tax savings that accrue to the firm from depreciation and amortization expense.

⁴ Net property, plant, and equipment is equal to the difference in the accumulated cost of all property, plant, and equipment (gross PPE) less the accumulated depreciation for those assets.

⁵ To predict CAPEX using this relationship requires that we estimate the change in net PPE for 2010 and depreciation expense. Typically, the former involves relating changes in net PPE and predicted changes in project revenues.

⁶ A distinction is sometimes made between depreciation as the cost of maintenance of existing assets and CAPEX, which represents expenditures on capital and equipment required to support anticipated growth.

growing revenues, which require added plant and equipment capacity. For example, when a firm anticipates growth opportunities, it spends more on long-lived assets than the amount it depreciates on older assets. The amount of CAPEX may also exceed depreciation expense if the cost of new assets is rising or if existing assets are depreciated at a rate that is slower than the actual rate at which the assets physically deteriorate. Similarly, there are situations in which CAPEX may be less than depreciation expense. Thus, in general, we should not expect CAPEX to equal the firm's depreciation expense.

Changes in net working capital (WC) Just as the firm must invest in property, plant, and equipment as it grows, it must also invest in current assets such as inventories and accounts receivable.⁷ However, the additional investment in inventories and receivables is partially financed by increases in the firm's trade credit that arise naturally in the course of the firm's purchases from its suppliers. The end result is that the firm incurs an outlay for working capital equal to what we will refer to as the change in **operating net working capital**. We define operating net working capital as follows:

$$\begin{aligned} \text{Operating Net Working Capital} = & \left[\left(\text{Current Assets} \right) - \left(\text{Cash and Marketable Securities} \right) \right] \\ & - \left[\left(\text{Current Liabilities} \right) - \left(\text{Current Portion of Interest-Bearing Debt/Notes} \right) \right] \end{aligned}$$

The cash flow impact of any additional investment the firm must make in working capital is then measured by the *change* in operating net working capital. For any arbitrary year t , the change in operating net working capital is calculated as follows:

$$\text{Change in Operating Net Working Capital}_t = \left(\text{Operating Net Working Capital} \right)_t - \left(\text{Operating Net Working Capital} \right)_{t-1} \quad (2)$$

We used *change* in operating net working capital rather than *increase* because the change can be both positive and negative. For example, as a firm's or project's revenues grow, it will usually have to invest more in working capital, such that the change will be positive (representing a cash outflow). However, as a project winds down and revenues stabilize and then decline, the firm's need for working capital will decrease, which means that the change in operating net working capital becomes negative. When this happens, the change actually results in a net inflow of cash.

Note that failure to consider an investment's need for additional working capital and the funds required to finance those needs will result in an overestimate of the value of the

⁷ This can also include cash and marketable securities where the growth requires that the firm hold more liquidity. However, this is typically not as important as inventories required to support additional sales revenues and the accounts receivable that arise naturally out of credit sales.

investment. This is especially true for high-growth projects that require frequent infusions of added capital to finance the growing need for inventories and accounts receivable.

Example—project free cash flow (Project FCF) To illustrate the calculation of Project FCF, we will look at the investment being considered by JC Crawford Enterprises, a privately held firm located in the suburbs of St. Louis, Missouri. The firm is a holding company for six franchise businesses, each selling and supporting a different franchise. JC Crawford has grown rapidly from revenues of \$16.5 million and 101 employees in 2002 to more than \$150 million and 500 employees today.

In the fall of 2009, JC Crawford's management was considering the construction of a regional distribution center for its franchise business devoted to appliance repairs, Mr. Fix-It-Up. There are 50 Mr. Fix-It-Up franchisees located in and around Miami, Florida, and the regional distribution center is needed to reduce the time and cost of supplying the franchisees throughout southern Florida.

Panel a of Table 1 lists a total of 12 assumptions and predictions that underlie JC Crawford's cash flow calculations for the regional distribution center. We have simplified the analysis by aggregating various items in the income statement and balance sheet into a single line item (e.g., total operating expenses are combined into a single ratio of operating expenses—net of depreciation—to sales). In practice, analysts often include much more detail (i.e., more line items in the financial statements) than we have done in Table 1.

Panel b of Table 1 contains pro forma income statements for the life of the investment spanning 2010–2012. The revenues for 2010 are estimated to be \$1 million and are expected to grow at a rate of 10% per year. The proposed investment requires a total investment of \$650,000 in new assets, comprised of \$250,000 in current assets (accounts receivable plus inventories equal 25% of sales). Because the firm is able to finance \$150,000 of this total from accounts payable (15% of sales), the increased investment in operating working capital for 2010 is only \$100,000 = \$250,000 – \$150,000. In addition to working capital, the firm will need to invest \$400,000 in plant and equipment in 2010, which is 40% of projected revenues for 2010.

As the firm's revenues grow over time, it will have to invest more in plant and equipment (i.e., CAPEX). Specifically, the firm will need to invest an amount equal to the sum of depreciation expense for the period plus 40% of the increase in sales. Thus, in 2010, Crawford will have CAPEX equal to 40% of the \$100,000 in sales growth it anticipates for 2011 plus the depreciation expense for 2010, for a total of \$80,000 (\$40,000 for new plant and equipment needed to support the increase in sales plus another \$40,000 to replace depreciated plant and equipment). In addition, operating net working capital will increase by 10% of new sales for 2010, or \$10,000. The 10% figure comes from the difference in current assets, which are estimated to equal 25% of sales and current liabilities, which are 15% of sales. Finally, the firm pays taxes equal to 30% of its taxable income and depreciates plant and equipment over a 10-year life toward a zero salvage value using straight-line depreciation.⁸

⁸ Straight-line depreciation, you will recall, involves allocating an equal amount of the cost of the asset over its life. For example, the \$400,000 CAPEX made in 2009 will lead to depreciation expenses of \$40,000 per year over the next 10 years, such that it will have a zero salvage value at the end of its 10-year life. For tax purposes, the IRS allows firms to use an accelerated depreciation method called the Modified Accelerated Cost Recovery System (MACRS). However, for illustration purposes, we use the straight-line method.

Table 1 Estimating Project Free Cash Flows for the Proposed Warehouse Distribution Center**Panel a. Assumptions and Predictions**

Predicted sales for 2010	\$1,000,000
Sales growth rate for 2011–2012	10%
Gross profit margin = Gross profit/Sales	30%
Operating expenses (before depreciation)/Sales	20%
Tax rate	30%
Gross plant and equipment/Sales	40%
Accounts receivable and inventories/Sales	25%
Accounts payable plus accrued expenses/Sales	15%
Short-term notes payable/Total assets	0%
Long-term debt/Total assets	0%
Depreciable life of plant and equipment	10 years
Depreciation method	Straight line

Panel b. Pro Forma Income Statements

	2010	2011	2012
Sales	\$1,000,000	\$1,100,000	\$1,210,000
Cost of goods sold	(700,000)	(770,000)	(847,000)
Gross profit	300,000	330,000	363,000
Operating expenses (before depreciation)	(200,000)	(220,000)	(242,000)
Depreciation expense	(40,000)	(50,000)	(61,000)
Earnings before interest and taxes	60,000	60,000	60,000
Interest expense	—	—	—
Earnings before taxes	60,000	60,000	60,000
Taxes	(18,000)	(18,000)	(18,000)
Net income	\$ 42,000	\$ 42,000	\$ 42,000

Panel c. Pro Forma Balance Sheet

	2010	2011	2012
Accounts receivable plus inventories	\$250,000	\$275,000	\$302,500
Gross plant and equipment	400,000	440,000	484,000
Less: Accumulated depreciation expense	(40,000)	(90,000)	(111,000)
Net plant and equipment	360,000	350,000	373,000
Total assets	\$610,000	\$625,000	\$675,500
Accounts payable plus accrued expenses	\$150,000	\$165,000	\$181,500
Short-term notes payable	—	—	—
Current liabilities	150,000	165,000	181,500
Long-term debt	—	—	—
Common equity	460,000	460,000	494,000
Total liabilities plus common equity	\$610,000	\$625,000	\$675,500

Table 1 *continued*

Panel d. Project Free Cash Flows (FCF)				
	2009	2010	2011	2012
Earnings before interest and taxes	—	\$60,000	\$ 60,000	\$ 60,000
Taxes	—	(18,000)	(18,000)	(18,000)
Net operating profit after taxes (NOPAT)	—	42,000	42,000	42,000
Plus: Depreciation expense (DA)	—	40,000	50,000	61,000
Less: Capital expenditures (CAPEX)	(400,000)	(80,000)	(94,000)	423,000
Less: Change in operating net working capital	(100,000)	(10,000)	(11,000)	121,000
Free cash flow (FCF)	<u>\$ (500,000)</u>	<u>\$ (8,000)</u>	<u>\$ (13,000)</u>	<u>\$647,000</u>
Legend:				
Analysis of the Investment in Operating Net Working Capital				
	2009	2010	2011	2012
Operating net working capital	—	\$ 100,000	\$ 110,000	\$ 121,000
Change in operating net working capital	(100,000)	(10,000)	(11,000)	121,000
Analysis of Net Property and Equipment Needs				
	2009	2010	2011	2012
Sales	—	\$1,000,000	\$1,100,000	\$1,210,000
Net property plant and equipment (40% of sales for next year)	400,000	440,000	484,000	—
Calculation of Capital Expenditures (CAPEX)				
	2009	2010	2011	2012
Increase in net property, plant, and equipment	\$400,000	\$ 40,000	\$ 44,000	\$ (484,000)
Plus: Depreciation expense	—	40,000	50,000	61,000
Capital expenditures	<u>\$400,000</u>	<u>\$ 80,000</u>	<u>\$ 94,000</u>	<u>\$ (423,000)</u>

Panel d contains the computation of the project FCFs for 2009 through 2012. Note that the project is initiated at the end of 2009 (beginning of 2010) and that we assume that the operating net working capital and net plant and equipment expenditures are made immediately. The project FCF for 2009 equals (\$500,000), which equals the \$400,000 spent on plant and equipment (40% of sales forecast for 2010) plus operating net working capital equal to 10% of the sales forecast for 2010. For 2010, both CAPEX and the new investment in operating net working capital are based on the increase in firm sales for 2011 over 2010 because the initial investments made for these two classes of assets remain in place from 2009 (except for depreciation expense, which is added to 40% of the change in sales for the year to estimate CAPEX).

Based on the three-year project life used in this analysis, the project FCFs are negative for 2009 through 2011, but the sale of plant and equipment and operating net working capital (assumed to be for its book value) produces a \$647,000 cash flow in 2012. Of course, the distribution center would probably have a much longer useful life than the three years used here for illustration purposes. It's now time to investigate how we might use the project FCF information to calculate value of the investment.

3 COMPREHENSIVE EXAMPLE—FORECASTING PROJECT FREE CASH FLOWS

Forecasting cash flows for prospective investments is not an exact science but is, rather, a mixture of science, intuition, and experience. To illustrate the techniques that firms can use to make cash flow forecasts, we analyze an investment involving a high-tech product—liquid crystal display (LCD) panels for big-screen television sets.

By the close of 2003, it was obvious that flat-panel wall-hanging TVs were fast becoming the preferred format for high-end TVs. Plasma-screen models were the most popular technology for monitors larger than 30 inches diagonally. However, when plasma screens are compared to LCDs, they have some rather serious limitations, including a shorter life span (30,000 hours compared to 50,000) and screen “burn in.” With plasma screens, if the TV is left on with a channel logo or other fixed image displayed for many hours, “burned-in” images sometimes appear as shadows on the screen. The LCDs, on the other hand, tend to have superior resolution in daylight hours and do not suffer from burn-in problems. However, the technology for making LCDs larger than 30 inches was only starting to be developed.

By the end of 2003, Lecion Electronics Corporation had been involved in the manufacture of flat-panel LCDs for use as computer monitors for 10 years. The company possessed some of the leading technology for both the design and manufacture of large-screen LCDs. The firm is currently considering the cash flow consequences of an investment of \$200 million in a new fabrication plant in South Korea that can produce 42-inch LCDs over the next five years using the firm’s proprietary technology. The investment would almost overnight make Lecion a major supplier of large-screen LCDs for sale in the home-entertainment market. Lecion’s marketing group believes that the firm can capture an estimated 20% market share once the plant is in full production.

Lecion’s Strategic Assessment of the Opportunity

Very early in the project evaluation analysis, Lecion Electronics Corporation engaged in a thorough assessment of the proposed strategy. This **strategic assessment** process centers on “the story” that underlies the investment. Specifically, they need to answer one overarching question: What are the specific capabilities of the firm and the competitive circumstances that will allow it to realize this positive-NPV opportunity?

In trying to address this question, it helps to have a systematic approach that incorporates all the fundamental questions that a firm should answer each time it contemplates a project. One critical aspect of a strategic assessment is that we *explicitly state* the assumptions underlying the investment plan. Lecion’s project management team addressed the potential strategic implications of the new investment (using the strategic assessment questions set forth). They came to the following conclusions:

- *Assumptions regarding competitor responses.* Four major manufacturers have the technological expertise and capacity to compete in the large flat-screen market. Each is expected to continue its efforts to compete for market share. However, the recession following the meltdown of the dot-com market in 2000 led the manufacturers to cut back on their development programs; Lecion has taken a clear technological lead that is expected to hold up for at least a year and possibly two. Furthermore, Lecion’s management feels that its technological edge allows the firm to capture a position of leadership in the industry over the first 12 months of

production. The market share the firm anticipates capturing during this period will help it weather any storm of competitor products over the five-year projected life of the investment project being contemplated.

- *Assumptions regarding producers of complementary products.* Lecion has worked closely with producers of high-end audio products to ensure the compatibility of its LCDs in all their newest product offerings. The firm continues to monitor developments in new wireless technology that will greatly reduce the cost of installing home-theater systems. The firm expects to grow its share of the home-theater market as its LCDs grow in popularity.
- *Assumptions regarding customer responses.* The early demand for the 42-inch LCDs is expected to come from customers in the upper tiers of consumable income who are not expected to be particularly price-sensitive. This will eventually change over the five-year life of the project, and the firm's estimate of the price per unit of its LCDs incorporates consideration for this declining product price.
- *Assumptions regarding how employees will respond.* With the economy having moved out of a recessionary period and employment levels growing, Lecion expects that the job market will become increasingly more competitive and tight. As a consequence, they anticipate having to pay higher wages to their employees. However, because the plant is located in South Korea, where wage costs are substantially lower than in the United States, the pressures of wage inflation are not considered to be a significant factor driving the economics of the new plant investment.

The key point to make before we embark on our analysis of “the numbers” is that it must rely upon and be consistent with the strategic assessment story that underlies the investment. If the story is inconsistent with the forecasts that are the basis for the cash flow estimates, then the analysis is fatally flawed. The strategic analysis process is necessarily subjective and appears unsophisticated in comparison with the apparent precision of the cash flow numbers. However, it is crucial that the story and the numbers build upon one another if the project analysis is to produce the information required to make a good decision about the project.

Estimating the Investment's Project Free Cash Flow (PFCF)

Our earlier description of the calculation of Project FCF for any period t can be restated in equation form as follows:

$$\begin{aligned} \text{FCF}_t = & \left[\left(\frac{\text{Firm}}{\text{Revenues}_t} \right) - \left(\frac{\text{Cost of}}{\text{Goods Sold}_t} \right) - \left(\frac{\text{Operating}}{\text{Expenses}_t} \right) - \left(\frac{\text{Depreciation}}{\text{Expense}_t} \right) \right] \\ & \times \left(1 - \frac{\text{Tax}}{\text{Rate}} \right) + \left(\frac{\text{Depreciation}}{\text{Expense}_t} \right) - \text{CAPEX}_t - \frac{\text{Change in}}{\text{Working Capital}_t} \quad (3) \end{aligned}$$

Viewed in this way, it is apparent that the firm's cash flow forecasting problem involves developing estimates for *each* of these components of FCF for *each year* of the project's anticipated life. Throughout our discussion, we will refer to these key components as value drivers because they determine the investment's value.

Forecasting Incremental Revenues

The first value driver found on the right-hand side of Equation 3 is firm revenues. Here we refer to the incremental revenues that result from undertaking the LCD investment.

Because the LCD investment will provide revenues only if the investment is undertaken, all of the plant's revenues are incremental to Lecion and are therefore relevant to our analysis.

To develop our forecast of revenues from the investment in the new LCD fabrication plant (or simply "fab"), we begin by defining the key cash flow drivers for firm revenues. To keep matters as uncluttered as possible, we will assume that Lecion's new plant will produce a single product (42-inch LCD panels). This means that the plant's revenues for any given period are equal to the product of the volume of units sold by the firm times the market price received for each unit sold. Additionally, it is often helpful to think of the firm's units produced and sold as a fraction of the total number of units of the product sold by all firms.

In our example, Lecion's revenues for the period are equal to the product of total unit sales for all firms in the industry times Lecion's projected market share multiplied by the market price per unit. Thus, we define Lecion's revenues for period t as follows:

$$\text{Lecion's Revenues}_t = \left(\frac{\text{Total Units Sold}_t}{\text{Units Sold}_t} \right) \times \left(\frac{\text{Lecion's Market Share}_t}{\text{Market Share}_t} \right) \times \left(\frac{\text{Market Price per Unit}_t}{\text{Unit}_t} \right) \quad (4)$$

Table 2 contains projected revenues for Lecion's LCD plant, spanning the five years beginning in 2004. To generate the forecast of plant revenues, we need to estimate the total units of 42-inch LCDs sold by all manufacturers, the firm's share of those total units, and the price per unit. Although we do not present the details underlying these projections here, the estimates resulted in the set of market demand numbers found in the second column of Table 2.

Next, the project management team estimates that Lecion will maintain a market share of 20% of total industry sales of 42-inch LCDs in each of the next five years. Obviously, this fraction is subject to a high level of uncertainty, and some type of sensitivity analysis on this variable would be appropriate.

The final element of the revenue forecast is the price forecast for the 42-inch LCDs. The team predicts that the price of these panels will be \$8,000 in 2004 and that this price will drop to \$4,098 by the end of 2008. This forecast reflects the use of a *price decay function*, which is commonly used in evaluating prices of high-tech products such as computer semiconductors. In this instance, the decay function assumes a rate of price decay of 20%. This means that every time the cumulative market volume of LCDs produced and sold doubles, the price of an LCD drops by 20%. This relationship is sometimes referred to as the *experience curve*. This functional relationship between product price and cumulative

Table 2 Market Demand and Revenue Projections for Lecion's LCD Fabricating Plant

Year	Total Units Sold (Market)	Firm Units Sold (Total Units Sold × Market Share)	Price/Unit	Firm Revenues
2004	10,000,000	2,000,000	\$8,000	\$16,000,000,000
2005	15,000,000	3,000,000	5,959	17,877,676,307
2006	20,000,000	4,000,000	4,932	19,727,751,418
2007	20,000,000	4,000,000	4,381	17,525,478,548
2008	15,000,000	3,000,000	4,098	12,294,285,429

market units produced and sold has been observed over and over in a wide variety of high-tech products.⁹

Estimating Total Annual Expenses

Lecion's total annual expenses are comprised of cost of goods sold, cash operating expenses, and depreciation expenses for plant and equipment, i.e.,

$$\begin{array}{ccccccc} \text{Total} & & \text{Cost of Goods} & & \text{Cash Operating} & & \text{Depreciation} \\ \text{Expenses} & = & \text{Sold for} & + & \text{Expenses} & + & \text{Expense} \\ \text{for Year } t & & \text{Year } t & & \text{for Year } t & & \text{for Year } t \end{array}$$

To forecast these expenses, we combine cost of goods sold and operating expenses. The sum of these two expense categories is then decomposed into variable and fixed components. Thus, for forecasting purposes, we define the sum of the firm's annual cost of goods sold and operating expenses as follows:

$$\begin{aligned} \left(\begin{array}{cc} \text{Cost of Goods} & \text{Cash Operating} \\ \text{Sold for} & \text{Expenses for} \\ \text{Year } t & \text{Year } t \end{array} \right) &= \left(\begin{array}{cc} \text{Variable Cost} & \text{Lecion's Unit} \\ \text{per Unit} & \times \text{Sales in} \\ \text{in Year } t & \text{Year } t \end{array} \right) \\ &+ \text{Fixed Operating Costs in Year } t \end{aligned} \quad (5)$$

TECHNICAL INSIGHT

Salvage Values and Terminal Values

The terms *salvage value* and *terminal value* are often used interchangeably when referring to the final cash flow that is realized at the end of the period used to value an investment (i.e., the analysis period). However, we want to draw a subtle but meaningful distinction between these terms for purposes of discounted cash flow analysis. Specifically, when valuing individual projects that have a finite life, we use the term *salvage value* to refer to the disposal value of the project assets at the end of the term of our analysis. This value is typically what the term *salvage* connotes, in that we dispose of the project assets and the proceeds make up the terminal value of the investment.

However, in some very long-term investments where the productive life of the project assets is expected to continue indefinitely, the terminal value represents the present value of all investment cash flows beyond the end of the analysis period. When we evaluate the value of a firm, the terminal value estimate is critical, for it can often contain the majority of the value of the enterprise. For our purposes in this chapter and the next, however, we will assume that the analysis period is equal to the productive life of the investment, such that the terminal value of the project is actually the salvage value of the investment assets.

⁹ For example, Lee et al. (1997) observed that a 20% price decay function explained more than 88% of the variation in unit semiconductor prices for a major supplier of microprocessors during the early 1990s (Winyih Lee, John Martin, and Hirofumi Matsuo, "Valuing Investments That Reduce Time-to-Market in the Semiconductor Industry," *Advances in Financial Planning and Forecasting* 7 [1997]: 19–23).

Table 3 Variable and Fixed Costs per Unit Estimates for the Lecion LCD Plant

Year	Firm Units Sold (Total Units Sold × Market Share)	Variable Operating Cost per Unit	Variable Cash Operating Cost	Fixed Cash Operating Costs	Depreciation Expense	Total Expenses
2004	2,000,000	\$9,000	\$(18,000,000,000)	\$(50,000,000)	\$(40,000,000)	\$(18,090,000,000)
2005	3,000,000	6,238	(18,713,642,143)	(50,000,000)	(40,000,000)	(18,803,642,143)
2006	4,000,000	4,610	(18,438,849,764)	(50,000,000)	(40,000,000)	(18,528,849,764)
2007	4,000,000	3,815	(15,259,902,966)	(50,000,000)	(40,000,000)	(15,349,902,966)
2008	3,000,000	3,428	(10,285,091,068)	(50,000,000)	(40,000,000)	(10,375,091,068)

In Equation 5, we need to forecast variable cash operating cost per unit and the firm's total fixed cash operating expenses per year. (We have the other amounts from our previous discussion.) Lecion estimates the variable cost per unit in 2004 to be \$9,000 (see Table 3).

Lecion's analysts estimate that the variable cost per unit will decline by 30% each time the cumulative volume of production doubles. In this case, the cost estimate reflects a *70% learning curve model*.¹⁰ That is, each time production volume doubles, Lecion estimates that the per unit variable cost will drop to 70% of the variable cost per unit, corresponding to half the current volume. For example, if variable costs were \$10,000 for 500,000 units, then Lecion would expect them to drop by 30% to \$7,000 when the cumulative volume produced reached 1,000,000 units. Of course, there's nothing magical about the notion of a learning curve. It simply represents a tool for describing the effects of learning and continuous improvements on the manufacturing process over the life cycle of the product. Such learning, in turn, lowers the variable costs of production.

Lecion's analysts made four simplifying assumptions in compiling investment cash flow estimates: First, they assumed that no new investments in net working capital will be required over the life of the project. Second, they assumed that plant and equipment will be depreciated using straight-line depreciation, resulting in a zero estimated salvage value. Third, the plant is analyzed as if it had no continuing or terminal value. This is probably an overly conservative assumption, because the plant is likely to have some value either as scrap or possibly as a plant site for a later technology fab. Nonetheless, for purposes of the analysis carried out here, the estimated salvage value is zero. Finally, investment income is assumed to be taxed at a constant rate for all periods and all levels of income.

A review of the FCF estimates found in Table 4 indicates that Lecion's analysts expect the project to have a positive cash flow by the end of 2006 and to remain positive throughout the project's remaining life. Moreover, the contribution margin in the final column of Table 4 reveals a growing margin of profits over the project's life. (Contribution margin is defined as price per unit less variable cost per unit divided by price per unit.) These forecasts are, of course, predictions and subject to forecast error. For now, with the forecasts of the FCF in hand, we focus on the process of valuing Lecion's expected future cash flows as a next step.

¹⁰ Willyard and McClees (1987) provide some empirical evidence that in high-tech applications, the learning curve reflects about 78% (Charles H. Willyard and Cheryl W. McClees, "Motorola's Technology Roadmap Process," *Research Management* 30, no. 5 [September–October 1987]: 13–19).

Table 4 Estimated Project Free Cash Flow for the Lecion LCD Investment

Year	Revenues	Total Expenses	NOPAT	Depreciation	FCF	Contribution Margin
2004	\$16,000,000,000	\$(18,090,000,000)	\$(1,463,000,000)	\$40,000,000	\$(1,423,000,000)	−12.5%
2005	17,877,676,307	(18,803,642,143)	(648,176,085)	40,000,000	(608,176,085)	−4.7%
2006	19,727,751,418	(18,528,849,764)	839,231,158	40,000,000	879,231,158	6.5%
2007	17,525,478,548	(15,349,902,966)	1,522,902,908	40,000,000	1,562,902,908	12.9%
2008	12,294,285,429	(10,375,091,068)	1,343,436,053	40,000,000	1,383,436,053	16.3%

Legend:

NOPAT = (Revenues in Year t − Total Expenses in Year t)(1 − Tax Rate)

FCF in Year t = NOPAT in Year t + Depreciation Expense in Year t

Contribution Margin in Year t = $\frac{(\text{Price per Unit in Year } t - \text{Variable Cost per Unit in Year } t)}{\text{Price per Unit in Year } t}$

Note: There are no capital expenditures after 2003 and no investments in operating net working capital.

4 VALUING INVESTMENT CASH FLOWS

Once the analyst has estimated the future cash flows from an investment, it is time to value them. Very simply, we need to discount the forecasted future cash flows back to the present time using a discount rate that properly reflects the anticipated risks of the future cash flows.

Figure 1 summarized the DCF process. Up to this point, we have dealt *only* with Step 1, which involves estimating investment cash flows. For now, for convenience, we will assume that we have performed Step 2 and that we know the appropriate discount rate for use in calculating the present value of future cash flows.¹¹

Example—Valuing Lecion’s Project Cash Flows

To illustrate the procedure used to carry out Step 3 of the DCF valuation process, we will value the FCFs estimated for the Lecion LCD project found in Table 4. Below, we provide a quick review of the mathematics of discounting for those who need it. (See the discussion found in the Technical Insight box (later in this chapter) on the mechanics of time value of money calculations.)

The opportunity cost of capital for the Lecion project is 18% per year. The discount factor for each year’s cash flow is defined as follows:

$$\text{Discount Factor for year } t = \left(\frac{1}{1 + .18} \right)^t$$

The present value of each year’s cash flow is simply the product of the cash flow and a discount factor for the year. Summing the present values of the cash flows generated by the project in

¹¹ This refers to the proper discount rate and the calculation of the cost of capital.

INDUSTRY INSIGHT

Do corporate executives use DCF?

When asked whether they always or almost always use IRR, 75.7% of corporate CFOs responded in the affirmative. Similarly, 74.9% said they always or almost always used NPV. These results came from a survey done by John Graham and Campbell Harvey of 392 CFOs* and reveal a dramatic increase in the reliance on NPV compared to IRR. In a survey completed in 1977, fewer than 10% of the respondents relied on NPV as their primary method, while more than 50% said they relied primarily on IRR.**

*John Graham and Campbell Harvey, "The Theory and Practice of Corporate Finance: Evidence from the Field," *Journal of Financial Economics* 60 (2001).

**L. Gitman and J. Forrester, Jr., "A Survey of Capital Budgeting Techniques Used by Major U.S. Firms," *Financial Management* (1977): 66–71.

each year of the project's operations (Years 1 through 5, found in Table 4), Lecion estimates the value of the investment to be \$303,253,010.

Using NPV and IRR to Evaluate the Investment

If Lecion invests \$200 million in the project, as indicated in Table 5, then the project will produce \$103,253,010 more, in present value terms, than it costs.¹² This difference in the present value of the project's expected future cash flows and the initial cost of making the investment is commonly referred to as the **net present value**, or simply NPV.

Another popular indicator of the anticipated wealth created by an investment is the **internal rate of return**, or IRR, of the project. We define the IRR as the compound rate of return earned on the investment, and we calculate it by solving the following expression:

$$\text{Investment Outlay}_0 = \sum_{t=1}^N \frac{\text{FCF}_t}{(1 + \text{IRR})^t} \quad (6)$$

Table 5 Valuation and Discounting Cash Flows

Year	Estimated Project Free Cash Flow (FCF)	Discount Factor	Present Value
2003	\$(200,000,000)	1.0000	\$(200,000,000)
2004	(1,423,000,000)	0.8475	(1,205,932,203)
2005	(608,176,085)	0.7182	(436,782,595)
2006	879,231,158	0.6086	535,127,227
2007	1,562,902,908	0.5158	806,127,933
2008	1,383,436,053	0.4371	604,712,649
Project intrinsic value			\$303,253,010
Initial capital expenditure in 2003			\$(200,000,000)
Net present value			\$103,253,010

¹² For simplicity, we have assumed that the total investment in the project is made immediately. Obviously, this cannot be the case, as it would take some period of time for new plant facilities to be constructed and equipment to be ordered and put into operation. Furthermore, firms typically "stage" investments of the magnitude considered in this book. For example, they might commit the minimum amount of money required to capture the investment opportunity and begin the project, but reserve judgment on successive stages of investment until certain milestones have been reached.

Solving Equation 6 for the IRR, Lecion's analysts get 20.24%.¹³ Comparing the 20.24% IRR to the 18% cost of capital we used as our discount rate in calculating the value of the investment, Lecion concludes that the investment should be viewed favorably.

Both the NPV and IRR of the investment indicate that the investment creates value. Does this imply that Lecion should commit to the project and invest \$200 million over the coming year? In a world of certainty where forecasts are made without error, the answer is an unqualified yes. However, in this particular case, the analysis is based on very uncertain cash flow estimates. When uncertainty is present, there will be estimation error, and this estimation error can lead to systematic errors in decision making.

It is possible that overconfident managers have provided optimistic forecasts rather than true expected cash flows. Alternatively, the cash flow numbers may have been generated by managers with a vested interest in getting the project approved. As a consequence, firms generally will not accept projects simply because they have positive NPVs or because they have IRRs that exceed the appropriate cost of capital. For these reasons, it is important to dig deeper to understand the uncertainties that underlie both the assumptions and forecasts made in generating the NPV estimate before making the final decision.

Mutually Exclusive Projects

Up to this point in our analysis, we have been evaluating a single project or investment. Often, the analysis will entail consideration of multiple alternatives or competing projects, where the firm must select only one. We refer to these alternatives as **mutually exclusive investments** because the selection of one precludes investment in the others.

When Duke Energy is faced with the opportunity to construct a new power plant, the firm has a number of choices it can make regarding the fuel that is to be used in firing the plant (e.g., natural gas, fuel oil, nuclear, coal, or some mixture). Each technology has its particular advantages and disadvantages, but ultimately Duke must select one and only one. When ranking the alternatives, the NPV still provides the best metric, as it measures the expected contribution of the project to the value of Duke Energy. However, in some cases, firms need to choose between competing (mutually exclusive) investments because

Did you know?

Excel's NPV function does not calculate net present value.

Excel includes an NPV function with the following arguments:

$$\text{NPV}(\text{rate}, \text{value1}, \text{value2}, \dots)$$

where rate is the discount rate and value1, value2, and so forth are cash flows received 1, 2, and more periods in the future.

Consequently, the NPV function calculates the present value of a stream of future cash flows, with the first being received one period hence. Net present value, on the other hand, incorporates consideration for an initial outlay in the current period that is not discounted at all. Therefore, we can use the NPV function to calculate net present value as follows:

$$\begin{aligned} \text{Net Present Value} &= \text{NPV}(\text{rate}, \text{value1}, \text{value2}, \dots) \\ &\quad - \text{Initial Cash Outlay}_0 \end{aligned}$$

¹³ Solving for IRR can be quite laborious if done without the help of a calculator or spreadsheet. In Excel, there is a built-in IRR function that can be used, and almost all business calculators have a similar feature. If you are relying on these tools, however, you should be aware that multiple IRRs can and often do exist and that these built-in functions will not tell you that there is more than one IRR. See the Technical Insight box on next page on multiple internal rates of return for a discussion of this issue.

TECHNICAL INSIGHT

Multiple Internal Rates of Return (IRRs)

The IRR is defined simply as the compound annual rate of return earned on an investment. In introductory analyses of investment projects, we generally assume that the negative cash flows associated with a project, such as the development costs, come early in the project's life and that the positive cash flows come later. In reality, it is often the case that positive and negative cash flows are interspersed. When this is the case, there can be multiple rates of return, or IRRs, that make the NPV of the investment zero.

We can use the net present value (NPV) profile (a graph of NPV for various discount rates) to illustrate the problem. First, consider a typical investment with the following cash flow pattern:

Year 0	Year 1	Year 2	Year 3
\$(500)	\$400	\$300	\$50

The IRR for these cash flows is 31.44%, which is readily identifiable in the project's NPV profile, found at right:

Now consider the following set of project cash flows:

Year 0	Year 1	Year 2	Year 3
\$(200)	\$1,200	\$(2,200)	\$1,200

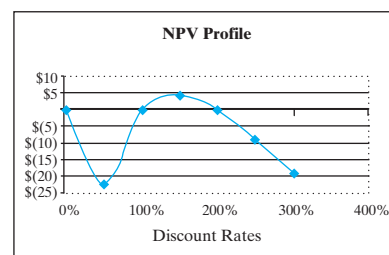
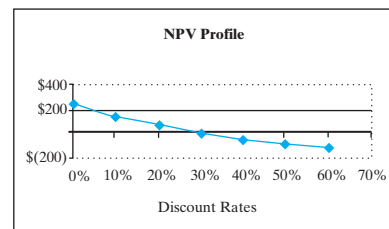
Note that these cash flows change sign from year to year on three occasions. It turns out that this fact drives the result that there are up to three different IRRs (see right):

In this example, there are three IRRs that make the NPV equal to zero: 0%, 100%, and 200%.

There are instances where no IRR can be computed. We leave the illustration to the reader. Consider the following pattern of cash flows:

Year 0	Year 1	Year 2
\$100	\$(300)	\$350

What is your estimate of the IRR for this stream of cash flows?



INDUSTRY INSIGHT

The Payback Model

Although it is widely derided among academic scholars, the payback model is often used in corporate practice as a tool for evaluating new investments.* The method is straightforward and involves estimating the number of years of expected future cash flows that are required to sum to the investment's initial outlay. For example, a \$4 million investment today that produces annual cash flows of \$1 million per year over a seven-year period has a payback period of four years.

Payback has three well-known drawbacks. First, it does not account for the time value of money. Second, it ignores the value of cash flows received after the payback period. Finally, the cutoff is not tied to market conditions; it depends on the manager's biases and may often be outdated.

To respond to the first issue, some calculate a discounted payback by using cash flows that are discounted to account for the time value of money. For example, using a 10% discount rate and the earlier example, discounted payback would be calculated as follows:

	Years						
	1	2	3	4	5	6	7
Project free cash flow (FCF)	\$1,000,000	\$1,000,000	\$1,000,000	\$1,000,000	\$1,000,000	\$1,000,000	\$1,000,000
Present value of FCF	909,091	826,446	751,315	683,013	620,921	564,474	513,158
Cumulative present value	\$ 909,091	\$1,735,537	\$2,486,852	\$3,169,865	\$3,790,787	\$4,355,261	\$4,868,419
Discounted payback	—	—	—	—	—	5.63	—

The discounted payback is 5.63 years, rather than 4 years, when the time value of future cash flows is ignored.

Although it is never optimal to ignore future payouts, payback may provide a simple screen for project risk, because a quick payback means that the firm's investment is at risk for a shorter period of time. Along these lines, it is not uncommon for firms to adopt a screening standard for new projects that precludes consideration of projects with payback periods longer than some threshold, such as three years.

* Over 55% of the CFOs who responded to a study by John Graham and Campbell Harvey ("The Theory and Practice of Corporate Finance: Evidence from the Field," *Journal of Financial Economics* 60 [2001]) indicated that they either always or mostly always use payback when valuing new capital expenditure proposals.

TECHNICAL

INSIGHT

Quick Review of the Mechanics of
Time Value of Money Calculations

The majority of the projects that we analyze in this book are *long-lived* in the sense that they provide cash flows over many years. Thus, an important step in the valuation of such projects involves collapsing these future cash flows down to their present value so that we can compare that amount to the cost of making the investment. The present value of a future stream of cash flows is the equivalent sum in today's dollars to the set of future cash flows promised by the project.

To illustrate, consider a project that offers three years of future cash flows: C_1 , C_2 , and C_3 . If we assume that the cash flows are risk free, then the appropriate discount rate for the three cash flows is the rate of interest corresponding to one-, two-, and three-year risk-free securities. These rates can be “inferred” from market prices of one-, two-, and three-year risk-free bonds as follows:

$$P_{1\text{-Year}} = \frac{\text{Face}_{\text{Year } 1}}{(1 + r_1)^1}, \quad P_{2\text{-Year}} = \frac{\text{Face}_{\text{Year } 2}}{(1 + r_2)^2}, \quad \text{and} \quad P_{3\text{-Year}} = \frac{\text{Face}_{\text{Year } 3}}{(1 + r_3)^3}$$

where P represents the current market values of the one-, two-, or three-year discount bond; r represents the current market rates of interest on the bonds; and *Face* is the maturity or face value of the bond that is paid to the holder at the end of each year. Note that the discount bonds pay no interest but simply return their full principal amount (Face) at maturity.

Using the risk-free rates for years one through three, we find the value of the three-year investment by the following formula:

$$\text{PV} = \frac{C_1}{(1 + r_1)^1} + \frac{C_2}{(1 + r_2)^2} + \frac{C_3}{(1 + r_3)^3} \quad (\text{A.1})$$

their ability to finance new investments is limited. In this situation, the NPV criteria is not necessarily the best way to evaluate the choice between the investments, and the firm may also want to consider the rate of return, or IRR, of the investments as well as how quickly the projects generate cash that can be used to fund other projects.

Many investment opportunities provide their own mutually exclusive alternative in the form of delayed execution or implementation. For example, Lecion's LCD investment opportunity was analyzed as if it had to be implemented immediately or the opportunity would be lost. It is often the case, however, that the firm that possesses the technological advantage has some degree of flexibility as to how quickly it moves to capitalize on that advantage. If, for example, Lecion could delay initiation of the LCD fab for up to one year, the information gained about market acceptance for similar products and the rate of price decay could make the project even more valuable to Lecion. Consequently, the firm's analysts should consider not only alternative technologies but also delayed execution of the investment as mutually exclusive alternatives. But for now, it is critical that the analyst consider whether or not the firm indeed has a viable option to delay and, if so, what the

Note that we have specified a different discount rate for each annual cash flow. To simplify matters, and because we are dealing with estimates, it is customary to use the same discount rate, r , for all future periods. That is,

$$PV = \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \frac{C_3}{(1+r)^3} = \sum_{t=1}^3 \frac{C_t}{(1+r)^t} \quad (\text{A.2})$$

The single discount rate assumption is particularly convenient where we consider the problems associated with estimating the appropriate discount rate for risky projects.

A popular variant of Equation A.2 describes the present value of a stream of cash flows that grows at a constant rate g , such that $C_2 = C_1(1+g)^1$ and $C_3 = C_1(1+g)^2$. If g is less than r , then we can rewrite Equation A.2 as follows:

$$PV = \left(\frac{C_1}{r-g} \right) \left(1 - \frac{(1+g)^3}{(1+r)^3} \right) \quad (\text{A.3})$$

If the number of periods in which cash flows are received gets very large, then Equation A.3 is approximated by:

$$PV = \left(\frac{C_1}{r-g} \right) \quad (\text{A.4})$$

Equation A.4 is commonly referred to as the **Gordon growth formula** since it was originally specified in a paper by Myron Gordon.¹⁴

firm might learn by deferring the initiation of the investment that would make the valuation more concrete.

The problems encountered when analyzing the possibility of delaying project initiation provide an opportunity to emphasize an important fact of life that analysts face

¹⁴The Gordon growth model is another well-known DCF relationship in finance and represents the sum of the following geometric progression:

$$\sum_{t=1}^{\infty} \frac{C_0(1+g)^t}{(1+r)^t} = C_0 \sum_{t=1}^{\infty} \frac{(1+g)^t}{(1+r)^t},$$

which can be reduced to

$$C_0 \left(\frac{1+g}{r-g} \right) \quad \text{or} \quad C_1 \left(\frac{1}{r-g} \right),$$

where $r > g$, and C_1 is equal to $C_0(1+g)$.

daily. That is, no matter how sophisticated the tools that are brought to bear on the valuation problem, there is always a need to exercise judgment. The point of the analysis is to inform the analyst so that ultimately he or she can make a better decision.

5 SUMMARY

The value of an investment project is determined by the cash flows it produces. In this chapter, we have discussed valuing investment opportunities using discounted cash flow (DCF) analysis that incorporates the three-step process outlined below:

Steps	Project Valuation
Step 1: Forecast the amount and timing of future cash flows.	Forecast project free cash flow (FCF).
Step 2: Estimate a risk-appropriate discount rate.	Combine the debt and equity discount rate (weighted average cost of capital, WACC).
Step 3: Discount the cash flows.	Discount the project's FCF using the WACC to estimate the value of the project as a whole.

The main focus of this chapter is on identifying and forecasting the *incremental* revenues and costs that are required to estimate expected cash flows. Although cash flow analysis is a key element in investment valuation, the process used to make forecasts is necessarily subjective and fraught with potential estimation errors. Consequently, cash flow forecasting is part art and part science. This does not mean that the analyst should simply throw caution to the wind and engage in crystal-ball gazing, for good forecasting is grounded in sound economic analysis and rigorous attention to the details of proper cash flow definition.

A key learning point from this chapter is that forecasting is hard work and is subject to potentially very large errors. To minimize the effect of these forecast errors, good analysts follow a three-pronged approach: First, they take the forecasting problem very seriously and utilize all the information and technology they have at their disposal to arrive at their forecast. Second, they do post-forecast risk analysis, which helps them prepare for a wide range of possible outcomes. Finally, to the extent possible, they try to maintain flexibility in how the investment is implemented so that they can respond to unforeseen future events.

PROBLEMS

1 CALCULATING PRESENT VALUES Calculate the present value of each of the following cash flow streams, using a discount rate of 10%:

- \$500 received at the end of 5 years.
- \$500 received annually for each of the next 5 years.
- \$500 received annually for each of the next 50 years.
- \$500 received annually for 100 years.

2 CALCULATING THE INTERNAL RATE OF RETURN Singular Construction is evaluating whether to build a new distribution facility. The proposed investment will cost Singular \$4 million to construct and provide cash savings of \$500,000 per year over the next 10 years.

- a. What rate of return does the investment offer?
- b. If Singular were to invest another \$200,000 in the facility at the end of 5 years, it would extend the life of the project for 4 years, during which time it would continue receiving cash savings of \$500,000. What is the internal rate of return for this investment?

3 CALCULATING PROJECT FCF In the spring of 2010, Jemison Electric was considering an investment in a new distribution center. Jemison's CFO anticipates additional earnings before interest and taxes (EBIT) of \$100,000 for the first year of operation of the center in 2011, and, over the next five years, the firm estimates that this amount will grow at a rate of 5% per year. The distribution center will require an initial investment of \$400,000 that will be depreciated over a five-year period toward a zero salvage value using straight-line depreciation of \$80,000 per year. Furthermore, Jemison expects to invest an amount equal to the firm's annual depreciation expense to maintain the physical plant. These additional capital expenditures will also be depreciated over a period of five years toward a zero salvage value. Jemison's CFO estimates that the distribution center will need additional net working capital equal to 20% of new EBIT (i.e., the change in EBIT from year to year).

Assuming the firm faces a 30% tax rate, calculate the project's annual project free cash flow (FCF) for each of the next five years.

4 COMPREHENSIVE PROJECT FCF TCM Petroleum is an integrated oil company headquartered in Fort Worth, Texas. Income statements for 2009 and 2010 are found below (\$ millions):

	Dec. 2009	Dec. 2010
Sales	\$13,368.00	\$12,211.00
Cost of goods sold	(10,591.00)	(9,755.00)
Gross profit	2,777.00	2,456.00
Selling, general, & administrative expense	(698.00)	(704.00)
Operating income before depreciation	2,079.00	1,752.00
Depreciation, depletion, and amortization	(871.00)	(794.00)
Operating profit	1,208.00	958.00
Interest expense	(295.00)	(265.00)
Nonoperating income or expense	151.00	139.00
Special items		20.00
Pretax income	1,064.00	852.00
Taxes	(425.60)	(340.80)
Net income	\$ 638.40	\$ 511.20

In 2009, TCM made capital expenditures of \$875 million, followed by \$1,322 million in 2010. TCM also invested an additional \$102 million in net working capital in 2009, followed by a decrease in its investment in net working capital of \$430 million in 2010.

- a. Calculate TCM's FCF for 2009 and 2010. TCM's tax rate is 40%.
- b. Estimate TCM's FCF for 2012–2016 using the following assumptions: Operating income continues to grow at 10% per year over the next five years, CAPEX is expected to be \$1,000 million per year, new investments in net working capital are expected to be \$100 million per year, and depreciation expense equals the prior year's total plus 10% of the prior year's CAPEX. Note that because TCM is a going concern, we need not be concerned about the liquidation value of the firm's assets at the end of 2016.

5 INTRODUCTORY PROJECT VALUATION Steve's Sub Stop (Steve's) is considering investing in toaster ovens for each of its 120 stores located in the southwestern United States. The high-capacity conveyor toaster ovens, manufactured by Lincoln, will require an initial investment of \$15,000 per store plus \$1,500 in installation costs, for a total investment of \$1,860,000. The new capital (including the costs for installation) will be depreciated over five years using straight-line depreciation toward a zero salvage value. In addition, Steve's will also incur additional maintenance expenses totaling \$120,000 per year to maintain the ovens. At present, firm revenues for the 120 stores total \$9,000,000, and the company estimates that adding the toaster feature will increase revenues by 10%.

- a. If Steve's faces a 30% tax rate, what expected project FCFs for each of the next five years will result from the investment in toaster ovens?
- b. If Steve's uses a 9% discount rate to analyze its investments in its stores, what is the project's NPV? Should the project be accepted?

6 INTRODUCTORY PROJECT VALUATION South Tel Communications is considering the purchase of a new software management system. The system is called B-Image, and it is expected to drastically reduce the amount of time that company technicians spend installing new software. South Tel's technicians currently spend 6,000 hours per year on installations, which cost South Tel \$25 per hour. The owners of the B-Image system claim that their software can reduce time on task by at least 25%. The system requires an initial investment of \$55,000 and an additional investment of \$10,000 for technician training on the new system. Annual upgrades will cost the firm \$15,000 per year. Because the investment is comprised of software, it can be fully expensed in the year of the expenditure (i.e., no depreciation). South Tel faces a 30% tax rate and uses a 9% cost of capital to evaluate projects of this type.

- a. Assuming that South Tel has sufficient taxable income from other projects so that it can immediately expense the cost of the software, what are the free cash flows for the project for years zero through five?
- b. Calculate the NPV and IRR for the project.

7 INTRODUCTORY PROJECT VALUATION CT Computers Inc. is considering whether to begin offering customers the option to have their old personal computers recycled when they purchase new systems. The recycling system would require CT to invest \$600,000 in the grinders and magnets used in the recycling process. The company estimates that for each system it recycles, it would generate \$1.50 in incremental revenues from the sale of scrap metal and plastics. The machinery has a five-year useful life and

will be depreciated using straight-line depreciation toward a zero salvage value. CT estimates that in the first year of the recycling investment, it could recycle 100,000 PCs and that this number will grow by 25% per year over the remaining four-year life of the recycling equipment. CT uses a 15% discount rate to analyze capital expenditures and pays taxes equal to 30%.

- a. What are the project cash flows? You can assume that the recycled PCs cost CT nothing.
- b. Calculate the NPV and IRR for the recycling investment opportunity. Is the investment a good one based on these cash flow estimates?
- c. Is the investment still a good one if the Year 1 units recycled are only 75,000?
- d. Redo your analysis for a scenario in which CT incurs a cost of \$0.20 per unit to dispose of the toxic elements from the recycled computers. What is your recommendation under these circumstances?

8 PROJECT VALUATION Glentech Manufacturing is considering the purchase of an automated parts handler for the assembly and test area of its Phoenix, Arizona, plant. The handler will cost \$250,000 to purchase plus \$10,000 for installation and employee training. If the company undertakes the investment, it will automate a part of the semiconductor test area and reduce operating costs by \$70,000 per year for the next 10 years. However, five years into the investment's life, Glentech will have to spend an additional \$100,000 to update and refurbish the handler. The investment in the handler will be depreciated using straight-line depreciation over 10 years, and the refurbishing costs will be depreciated over the remaining five-year life of the handler (also using straight-line depreciation). In 10 years, the handler is expected to be worth \$5,000, although its book value will be zero. Glentech's tax rate is 30%, and its opportunity cost of capital is 12%.

Exhibit P8.1 contains cash flow calculations for the project that can be used in performing a DCF evaluation of its contribution to firm value. Answer each of the following questions concerning the project:

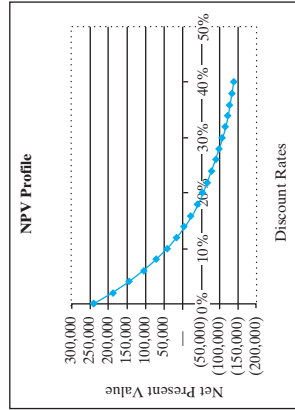
- a. Is this a good project for Glentech? Why or why not?
- b. What can we tell about the project from the NPV profile found in Exhibit P8.1?
- c. If the project were partially financed by borrowing, how would this affect the investment cash flows? How would borrowing a portion of the investment outlay affect the value of the investment to the firm?
- d. The project calls for two investments: one immediately and one at the end of Year 5. How much would Glentech earn on its investment, and how should we account for the additional investment outlay in our calculations?
- e. What are the considerations that make this investment somewhat risky, and how would you investigate the potential risks of this investment?

9 PROJECT VALUATION HMG is considering the manufacture of a new chemical compound that is used to make high-pressure plastic containers. An investment of \$4 million in plant and equipment is required. The firm estimates that the investment will have a five-year life, using straight-line depreciation toward a zero salvage value. However, the investment has an anticipated salvage value equal to 10% of its original cost.

The number of pounds of the chemical compound that HMG expects to sell over the five-year life of the project are as follows (in millions): 1, 1.5, 3, 3.5, and 2. To operate the new plant, HMG estimates that it will incur additional fixed cash operating expenses of \$1 million per year and variable operating expenses equal to 45% of revenues. Furthermore,

Exhibit P8.1**Glentech Manufacturing Company Cash Flow Estimates**

	0	1	2	3	4	5	6	7	8	9	10
Investment Outlays											
Equipment purchases	(250,000)					(100,000)					
Installation costs	(10,000)										
Initial outlay	(260,000)										
After-tax salvage value											3,500
Free Cash Flows											
Operating exp. savings		70,000	70,000	70,000	70,000	70,000	70,000	70,000	70,000	70,000	70,000
Less: Depreciation exp.		(26,000)	(26,000)	(26,000)	(26,000)	(26,000)	(46,000)	(46,000)	(46,000)	(46,000)	(46,000)
Added oper. income		44,000	44,000	44,000	44,000	44,000	24,000	24,000	24,000	24,000	24,000
Less: Taxes		(13,200)	(13,200)	(13,200)	(13,200)	(13,200)	(7,200)	(7,200)	(7,200)	(7,200)	(7,200)
NOPAT		30,800	30,800	30,800	30,800	30,800	16,800	16,800	16,800	16,800	16,800
Plus: Depreciation		26,000	26,000	26,000	26,000	26,000	46,000	46,000	46,000	46,000	46,000
Less: CAPEX	(260,000)	—	—	—	—	(100,000)	—	—	—	—	—
Free cash flow	(260,000)	56,800	56,800	56,800	56,800	(43,200)	62,800	62,800	62,800	62,800	66,300



HMG estimates that it will need to invest 10% of the anticipated increase in revenues each year in net working capital. The price per pound for the new compound is expected to be \$2 in Years 1 and 2, then \$2.50 per pound in Years 3 through 5. HMG's tax rate is 38%, and it requires a 15% rate of return on its new product investments.

Exhibit P9.1 contains projected cash flows for the entire life of the proposed investment. Note that investment cash flow is derived from the additional revenues and costs associated with the proposed investment.

- a. Does this project create shareholder value? How much? Should HMG undertake the investment?
- b. What if the estimate of the variable costs were to rise to 55%? Would this affect your decision?

Exhibit P9.1		HMG Project Analysis				
Givens:						
Investment	4,000,000					
Plant life	5					
Salvage value	400,000					
Variable cost %	45%					
Fixed operating cost	1,000,000					
Tax rate	38%					
Working capital	10%	of the change in revenues for the year				
Required rate of return	15%					
	0	1	2	3	4	5
Sales volume		1,000,000	1,500,000	3,000,000	3,500,000	2,000,000
Unit price		2.00	2.00	2.50	2.50	2.50
Revenues		2,000,000	3,000,000	7,500,000	8,750,000	5,000,000
Variable operating costs		(900,000)	(1,350,000)	(3,375,000)	(3,937,500)	(2,250,000)
Fixed operating costs		(1,000,000)	(1,000,000)	(1,000,000)	(1,000,000)	(1,000,000)
Depreciation expense		(800,000)	(800,000)	(800,000)	(800,000)	(800,000)
Net operating income		(700,000)	(150,000)	2,325,000	3,012,500	950,000
Less: Taxes		266,000	57,000	(883,500)	(1,144,750)	(361,000)
NOPAT		(434,000)	(93,000)	1,441,500	1,867,750	589,000
Plus: Depreciation		800,000	800,000	800,000	800,000	800,000
Less: CAPEX	(4,000,000)	—	—	—	—	248,000
Less: Working capital	(200,000)	(100,000)	(450,000)	(125,000)	375,000	500,000
Free cash flow	(4,200,000)	266,000	257,000	2,116,500	3,042,750	2,137,000
Net present value	419,435					
Internal rate of return	18.01%					

10 PROJECT VALUATION Carson Electronics is currently considering whether or not to acquire a new materials-handling machine for its manufacturing operations. The machine costs \$760,000 and will be depreciated using straight-line depreciation toward a zero salvage value over the next five years. During the life of the machine, no new capital expenditures or investments in working capital will be required. The new handler is expected to save Carson \$250,000 per year before taxes of 30%. Carson's CFO recently analyzed the firm's opportunity cost of capital and estimated it to be 9%.

- a.** What are the annual free cash flows for the project?
- b.** What are the project's net present value and internal rate of return? Is the project one that Carson should accept?
- c.** Carson's new head of manufacturing was concerned about whether the new handler could deliver the promised savings. In fact, he projected that the savings might be 20% lower than projected. What are the NPV and IRR for the project under this scenario?